

A two-dimensional counterexample to Conjecture 00000009761

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Abstract

The proposed arithmetic strengthening of the singular-value bound is false, already for a trace-class operator on $\mathbb{C}^2$. The upper triangular matrix $A = \begin{pmatrix} 1 & 3/2 \\ 0 & 1 \end{pmatrix}$ has eigenvalues $1, 1$ counted with algebraic multiplicity and singular values $2, 1/2$. At $n = 2$ the claimed upper bound is $3/4$, whereas $|\lambda_2| = 1$. We give an exact singular value decomposition, a Gram-matrix calculation, and a Lean 4 verification of the finite-dimensional counterexample.

1 Statement and conventions

The conjecture proposes the universal bound

$$|\lambda_n| \leq \frac{(\prod_{k=1}^n s_k)^{1/n} + s_n}{2}, \quad (1)$$

where the eigenvalues are listed in nonincreasing modulus, with algebraic multiplicity, and the singular values satisfy $s_1 \geq s_2 \geq \dots \geq 0$. The supplied English and Chinese statements contain no normality or self-adjointness hypothesis. The reference to second-order diagonal models concerns claimed attainment of optimality, not a restriction on the class of operators. We disprove the inequality itself, so its asserted universal optimality and uniqueness cannot hold as stated.

Proposition 1. *The trace-class operator on the standard complex Euclidean space $\mathbb{C}^2$ with matrix*

$$A = \begin{pmatrix} 1 & 3/2 \\ 0 & 1 \end{pmatrix}$$

violates (1) at $n = 2$.

Proof. Its characteristic polynomial is $(t-1)^2$, so $\lambda_1 = \lambda_2 = 1$. The adjoint is the conjugate transpose, and

$$A^*A = \begin{pmatrix} 1 & 3/2 \\ 3/2 & 13/4 \end{pmatrix}, \quad \det(tI - A^*A) = t^2 - \frac{17}{4}t + 1 = (t-4)(t-1/4).$$

Thus the ordered singular values, the nonnegative square roots of the eigenvalues of A^*A , are $s_1 = 2$ and $s_2 = 1/2$. All finite-dimensional operators are trace class; concretely, the sequence of singular values padded by zeros is $(2, 1/2, 0, \dots)$ and has sum $5/2$. At $n = 2$, the asserted right-hand side becomes

$$\frac{\sqrt{2 \cdot (1/2)} + 1/2}{2} = \frac{3}{4} < 1 = |\lambda_2|,$$

which contradicts (1). □

2 An explicit singular value decomposition

To independently certify that the listed values are singular values of the actual matrix, put $c = 1/\sqrt{5}$ and define

$$U = c \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \quad V = c \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}, \quad S = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

Direct multiplication gives $UU^* = VV^* = I$ and

$$USV^* = \frac{1}{5} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3/2 \\ 0 & 1 \end{pmatrix} = A.$$

Both diagonal entries of S are nonnegative and already in decreasing order. Hence this is a complete unitary SVD, with no numerical approximation. The ordinary product bound is not contradicted: for this example $|\lambda_1 \lambda_2| = s_1 s_2 = 1$. The additional averaging with the smaller singular value is the step that fails. The example is not a normal matrix; no assertion about a version restricted to normal operators is made here.

3 Formal verification and its scope

The project uses Lean 4.19.0 and a pinned Mathlib revision. The matrix in `lean/Main.lean` is an actual complex 2×2 Mathlib matrix. The kernel-checked statements establish:

- the actual characteristic polynomials and full root multisets of A and A^*A , including the repeated eigenvalue;
- membership of U, V in the actual complex unitary group, the exact identity $A = USV^*$, and the order and sign of the singular values;
- the standard Euclidean-space operator and the Mathlib identity identifying conjugate transpose with the Hilbert-space adjoint;
- summability of the verified singular values extended by zero, with exact sum $5/2$, and failure of the proposed $n = 2$ inequality.

`OrderedEigenvalues` is defined by the complete characteristic-root multiset and modulus order. `OrderedSingularValues` is defined by the existence of actual unitary matrices giving an SVD with a nonnegative, ordered diagonal. These are the standard finite-dimensional spectral data. The final theorem negates the universal inequality for these spectral data. The theorem `full_counterexample` collects the verified witness and its summable singular sequence. No general-purpose trace-class API or proof of Lidskii's theorem is claimed: the finite-dimensional trace-class bridge is precisely the summability criterion just stated. The $n = 2$ root is represented by the positive real square root.

The supplementary `verify.py` checks the matrix products, characteristic polynomials, and strict rational inequality using exact fractions. Fresh-build logs, a theorem-axiom audit, and the source hashes are provided in `verification/`. This proof uses no numerical search.

Source. The Last Math Competition, conjecture 00000009761, repository statement.

The submitted `SOURCE.md` preserves the bilingual snapshot byte for byte; its source identity is recorded in `verification/`.